

☒ HARD

☐ ADVANCED MATH

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# Advanced Math

# 03

10 questions · ~15 min

Q1

GRID-IN ADVANCED MATH

The expression  $6\sqrt[5]{3^5x^{45}} \cdot \sqrt[8]{2^8x}$  is equivalent to  $ax^b$ , where  $a$  and  $b$  are positive constants and  $x > 1$ . What is the value of  $a + b$ ?

STUDENT-PRODUCED RESPONSE

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

|   |   |   |   |
|---|---|---|---|
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A landscaper is designing a rectangular garden. The length of the garden is to be 5 feet longer than the width. If the area of the garden will be 104 square feet, what will be the length, in feet, of the garden?

## STUDENT-PRODUCED RESPONSE

|   |    |    |   |
|---|----|----|---|
|   |    |    |   |
| . | /. | /. | . |
| 0 | 0  | 0  | 0 |
| 1 | 1  | 1  | 1 |
| 2 | 2  | 2  | 2 |
| 3 | 3  | 3  | 3 |
| 4 | 4  | 4  | 4 |
| 5 | 5  | 5  | 5 |
| 6 | 6  | 6  | 6 |
| 7 | 7  | 7  | 7 |
| 8 | 8  | 8  | 8 |
| 9 | 9  | 9  | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

ADVANCED MATH

$$5(x + 7) = 15x - 17x + 7$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Which of the following expressions is(are) a factor of  $3x^2 + 20x - 63$ ?

I.  $x - 9$

II.  $3x - 7$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

The functions  $f$  and  $g$  are defined by the given equations, where  $x \geq 0$ . Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where  $x \geq 0$ ?

I.  $fx = 18(1.25)^x + 41$

II.  $gx = 90.73^x$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

$$\begin{aligned}x - y &= 1 \\ x + y &= x^2 - 3\end{aligned}$$

Which ordered pair is a solution to the system of equations above?

- A.  $(1 + \sqrt{3}, \sqrt{3})$
- B.  $(\sqrt{3}, -\sqrt{3})$
- C.  $(1 + \sqrt{5}, \sqrt{5})$
- D.  $(\sqrt{5}, -1 + \sqrt{5})$

Q7

ADVANCED MATH

Which of the following is equivalent to  $\left(a + \frac{b}{2}\right)^2$ ?

A.  $a^2 + \frac{b^2}{2}$

B.  $a^2 + \frac{b^2}{4}$

C.  $a^2 + \frac{ab}{2} + \frac{b^2}{2}$

D.  $a^2 + ab + \frac{b^2}{4}$

Q8

ADVANCED MATH

The function  $f$  is defined by  $f(x) = (-8)(2)^x + 22$ . What is the y-intercept of the graph of  $y = f(x)$  in the  $xy$ -plane?

A.  $(0, 14)$

B.  $(0, 2)$

C.  $(0, 22)$

D.  $(0, -8)$

Q9

ADVANCED MATH

Which quadratic equation has no real solutions?

A.  $x^2 + 14x - 49 = 0$

B.  $x^2 - 14x + 49 = 0$

C.  $5x^2 - 14x - 49 = 0$

D.  $5x^2 - 14x + 49 = 0$

Q10

ADVANCED MATH

$$f(x) = x^3 - 9x$$
$$g(x) = x^2 - 2x - 3$$

Which of the following expressions is equivalent to  $\frac{f(x)}{g(x)}$ , for  $x > 3$  ?

A.  $\frac{1}{x+1}$

B.  $\frac{x+3}{x+1}$

C.  $\frac{x(x-3)}{x+1}$

D.  $\frac{x(x+3)}{x+1}$